RIG-ETALE MAPS OF NOETHERIAN ADIC RINGS ARE STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof fixes a common ideal of definition, writes the six-term exact sequence of the cotangent complexes, transports the annihilation exponents from the two maps through the sequence, and applies the rig-etale equivalence lemma to the composite

Editorial note. The label lemma-composition-rig-etale-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the arrow property rig-etale for continuous homomorphisms of Noetherian adic rings and its meaning (lemma-rig-etale-continuous of the same chapter)
- the composition property of Formal Spaces, Remark formal-spaces-remark-composition-variant-Noetherian
- the exact sequence of completed naive cotangent complexes for A -> B -> C (lemma-exact-sequence-NL of the same chapter)
- the rig-etale criterion in terms of annihilation of the cohomology of NL^wedge by powers of an ideal of definition (lemma-equivalent-with-artin of the same chapter)

Source reference. Original remark 0GBK: sources/stacks-chow/tags.txt, source lines 20375–20375. Prerequisite pointer invoked at proof line(s) 4927; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJU: sources/stacks-restricted/pinned-original-source.tex, source lines 1810–1810. Prerequisite pointer invoked at proof line(s) 4960; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 4937; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCU: sources/stacks-restricted/pinned-original-source.tex, source lines 4749–4749. Prerequisite pointer invoked at proof line(s)

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4932; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The property $P(\varphi) =$"φ is rig-étale" on arrows of $WAdm^{Noeth}$ is stable under composition as defined in Formal Spaces, Remark 0GBK.*

Proof. The statement makes sense by Lemma 0GCU. To see that it is true assume we have rig-étale morphisms $A \to B$ and $B \to C$ in $WAdm^{Noeth}$. Then we can choose an ideal of definition $I \subset A$ such that the topology on C and B is the I-adic topology. By Lemma 00S2 we obtain an exact sequence

$$C \otimes_B H^0(NL^\wedge_{B/A}) \longrightarrow H^0(NL^\wedge_{C/A}) \longrightarrow H^0(NL^\wedge_{C/B}) \longrightarrow 0$$

$$H^{-1}(NL^\wedge_{B/A} \otimes_B C) \longrightarrow H^{-1}(NL^\wedge_{C/A}) \longrightarrow H^{-1}(NL^\wedge_{C/B})$$

There exists a $c \geq 0$ such that for all $a \in I$ multiplication by a^c is zero on $NL^\wedge_{B/A}$ in $D(B)$ and $NL^\wedge_{C/B}$ in $D(C)$. Then of course multiplication by a^c is zero on $NL^\wedge_{B/A} \otimes_B C$ in $D(C)$ too. Hence $H^0(NL^\wedge_{B/A}) \otimes_A C$, $H^0(NL^\wedge_{C/B})$, $H^{-1}(NL^\wedge_{B/A} \otimes_B C)$, and $H^{-1}(NL^\wedge_{C/B})$ are annihilated by a^c. From the exact sequence we obtain that multiplication by a^{2c} is zero on $H^0(NL^\wedge_{C/A})$ and $H^{-1}(NL^\wedge_{C/A})$. It follows from Lemma 0AJU that C is rig-étale over (A, I) as desired. □